Figure 1.4 shows the wooden pole structure used for LV transmission lines. The limitations of wooden poles are that they must be straight, strong with gradual tapper and free from knots. These poles are cheap. Some portion of the pole below the ground level is impregnated with preservative compounds like creosote oil. These poles are suitable for shorter spans between 40 m to 50 m and voltage level of 11 KV.

ii. Concrete poles



Fig. 1.5 Concrete pole

The concrete poles are strong and reinforced with iron rods and it is shown in figure 1.5. They have high mechanical strength, long life, durability and working conditions. These poles are used for longer spans (80 m to 100 m). The maintenance cost is very low. These types of poles are used to transmit up to 33 KV.

iii. Tubular steel poles

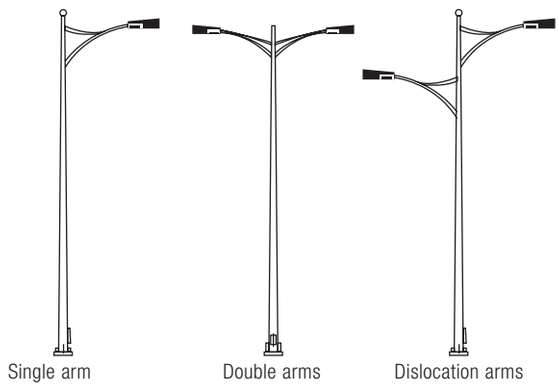


Fig. 1.6 Tubular poles

The iron tubular pipe is shown in Figure 1.6. This structure is stronger than the concrete pole. The poles are coated with zinc plating. In the same tube, more than one pipe is plugged or attached. Since the tube is circular shape, the air pressure attack is less than the concrete pole. These types of poles are required to be earthed. These are suitable for low voltage areas such as street lighting. This type of pole is suitable for the span between 50 m to 80 m.

iv. Latticed steel tower



Fig. 1.7 Latticed steel tower

A Latticed steel tower is shown in figure 1.7. These are designed to be of greater strength and longer life. The purpose of this tower is to carry a high voltage through the conductor to a long distance of high level. The span of the tower is 100m to 300m.

1.7 Effects of transmission

1.7.1 Skin effect

When an alternating current flows through a conductor, a flux will be produced in it. This flux will be higher at the center of the conductor than outer

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